

Assignment 1

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Student Name/Number: (doubleclick to edit)

Instructions

- Edit the above cell to include your name and student number.
- Submit just this `ipynb` to Canvas. Do not rename, it associates your student number with the submission automatically.

```
using Distributions, Plots, LaTeXStrings, LinearAlgebra, Statistics, Random
```

Question 1

Create the following variables:

- `D`: A floating point number with the value 10,000
- `r`: A floating point number with the value 0.025 (i.e., 2.5% net interest rate)

Compute the present discounted value of a payment (`D`) made in `t` years, assuming an interest rate of `r = 2.5%`. Save this value to a new variable called `PDV` and print your output.

Hint: The formula is

$$\text{PDV}(D, t) = \frac{D}{(1 + r)^t}$$

For $t = 10$, calculate this PDV

```
# add code here
```

Question 2

Now assume that you have an asset the pays D every year from $t = 0, \dots, T$. Write code which will price this as the PDV of all payoffs,

$$P_T(D) = \sum_{t=0}^T \left(\frac{1}{1+r} \right)^t D$$

Part (a)

Derive the analytic solution for the limit of $P_\infty(D) \equiv \lim_{T \rightarrow \infty} P_T(D)$

(doubleclick here to modify. Add other cells as required. No need to show all of your steps)

$$P_\infty(D) = ?$$

Part (b)

Plot the price as the horizon increases

- On the x-axis plot $T = 1, \dots, 30$
- On the y-axis plot $P_T(D)$ at that horizon
- Plot a horizontal line at the asymptotic P_∞ you calculated

```
T_max = 30
T = 1:T_max
r = 0.025
D = 10_000

# add code here
```

10000

Question 3

Now instead of having constant dividends, assume that dividends follow the process

$$\log D_{t+1} = \log D_t + \sigma w_{t+1}$$

Where

- $w_{t+1} \sim N(0, 1)$ are IID unit random normals
- $D_0 = 1.0$
- $\sigma = 0.001$

Part (a)

Write code to simulate a sequence of dividends with the process and initial condition for $t = 0, \dots, T = 30$.

```
T = 30
D_0 = 1.0
sigma = 0.001
# reminder, can draw from N(0,1) with randn()

# add code here for simulation
```

0.001

Part (b)

Plot three simulated sequences of dividends (i.e, the D_t for $t = 0, \dots, 30$) on the same graph with the shared x-axis.

```
# add code here for plotting using your functions above
```

Question 4

Using the simulated sequences of dividends from Question 3, calculate the P_T assuming perfect foresight (i.e., they were able to know the sequence of w_{t+1} even for $t \geq 0$). The formula remains the same, except where $\{D_0, \dots, D_T\}$ is an argument which allows for time-dependent dividends

$$P_T(\{D_t\}_{t=0}^T) = \sum_{t=0}^T \left(\frac{1}{1+r} \right)^t D_t$$

All from the same $D_0 = 1.0$ initial condition calculate the $P_T(\{D_t^n\}_{t=0}^T)$ for $n = 1, \dots, N$ simulated sequences of dividends (i.e. see Question 3)

Part (a)

Calculate the P_T above given a dividend sequence

```
T = 30
D_0 = 1.0
sigma = 0.001
# add code here for calculating P_T and check results
```

0.001

Part (b)

Plot a histogram of the prices for $N = 100$ simulations and compare to the deterministic case, which is nested if $\sigma = 0$. (Hint: see [our lectures](#) or [Julia By Example](#) for more on histograms)

```
N = 100
# add code here for plotting using your functions above
# hint: use the `histogram` function
```

100